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Course and Section	BIO 300 – A
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Instructors	Ma'am Lilibeth Bucol & Sir Abner Bucol
Species Name	<i>Anas flavirostris</i>

Genome Exploration II

Basic Genome Structure and Sequence Exploration Using Galaxy

Continuation Assignment • usegalaxy.org • Individual Work

Part 1 — Reconfirm the Genome Identity

Record the following in your report before starting the analysis:

Species name	<i>Anas flavirostris</i>
NCBI Assembly accession	GCA_038406055.1
Assembly level (if known)	Scaffold
FASTA filename in Galaxy	Anas flavirostris.fna.gz
Genome source/database	NCBI
Approximate file size	366.3 MB

The screenshot displays the Galaxy web interface. On the right, the 'History' panel shows a list of datasets generated from the uploaded 'Anas flavirostris_genome_original.fna.gz' file. The datasets include 'Anas flavirostris_Histogram', 'Anas flavirostris_Filtered_10kb_Assembly_Statistics_gfastats', 'Anas flavirostris_Filtered_10kb', 'Anas flavirostris_Compute sequence length (Sorted)', 'Anas flavirostris_Compute sequence length', 'Anas flavirostris_Assembly_Statistics_gfastats', and the original 'Anas flavirostris_genome_original.fna.gz'. The main workspace on the left shows a preview of the FASTA sequence, displaying the first 100 KB of the genome. The sequence is shown in a monospaced font, with line numbers on the left and the sequence data on the right. The sequence starts with '38CATY010000001.1 Anas flavirostris isolate UPM:Bird:24711 1, whole genome shotgun sequence'.

Figure 1. Galaxy History showing the uploaded *Anas flavirostris* genome dataset

Part 2 — Generate Basic Assembly Statistics

Record these values:

Metric	Values	Interpretation
Total assembly length	1,166,830,609	The assembly contains approximately 1.17 Gb of assembled DNA sequence, representing the total amount of sequence included in the genome assembly.
Number of sequences/scaffolds	194,180	The assembly contains 194,180 sequences , indicating that although several sequences are very long, the assembly also contains many smaller scaffolds.
Minimum sequence length	200	The shortest sequence is 200 bp , showing that the assembly contains very short sequences in addition to large scaffolds.
Maximum sequence length	201,470,136	The longest scaffold is 201,470,136 bp , indicating that a substantial portion of the assembly is represented by very long scaffolds.
Mean/average sequence length	6,009.02	The average sequence length is 6,009.02 bp , which is much smaller than the N50 because many short sequences are included in the assembly.
N50	75,318,589	The N50 of 75,318,589 bp means that 50% of the assembly is contained in sequences that are 75.3 Mb or longer .
L50	5	An L50 of 5 means that only the five longest sequences are sufficient to account for at least 50% of the total assembly length.
GC content (%)	41.74	The genome has a GC content of 41.74% , meaning approximately 41.74% of the assembled bases are guanine or cytosine.

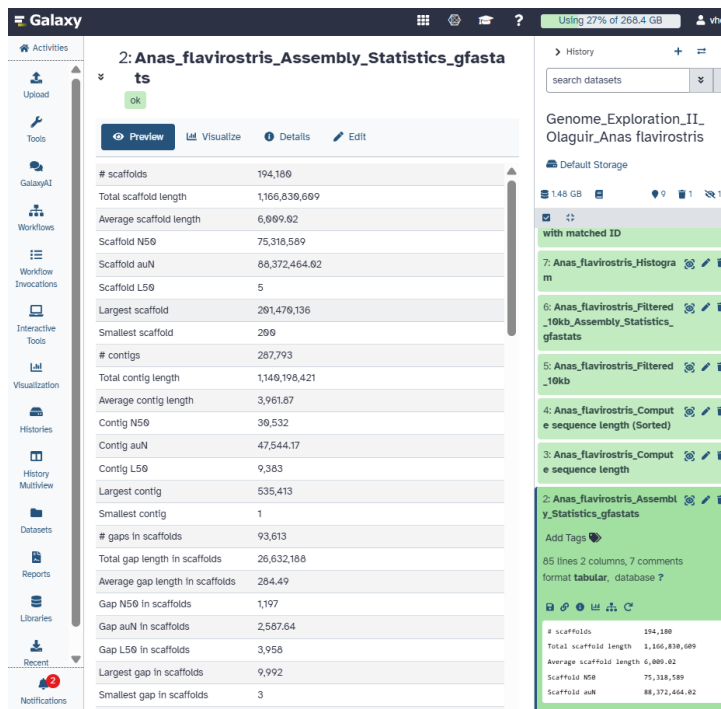


Figure 2. Galaxy gfastats output showing the assembly statistics of the *Anas flavirostris* genome.

Part 3 — Examine Sequence-Length Structure

Rank	Sequence ID	Length (bp)
1	JBCATY010000001.1	201470136
2	JBCATY010000047.1	154950756
3	JBCATY010000076.1	117403883
4	JBCATY010000086.1	79851999
5	JBCATY010000100.1	75318589

Optional visualization: if a plotting tool is available, make a histogram or bar chart of sequence lengths. This is optional because chromosome-level assemblies may contain only a small number of sequences.

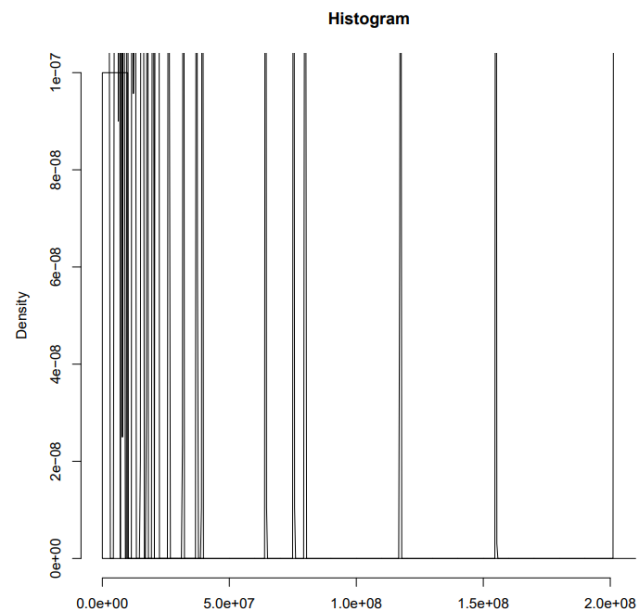
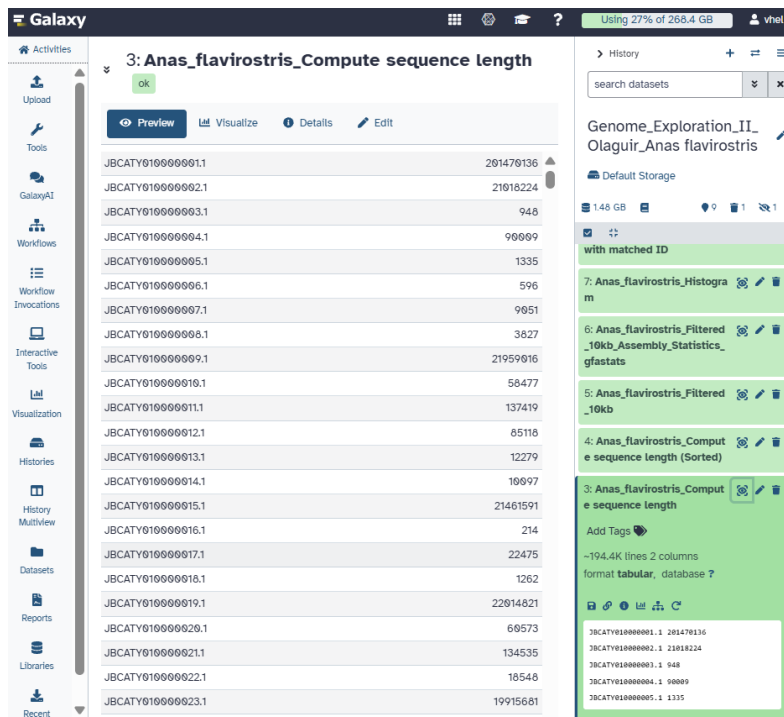


Figure 3. Galaxy histogram showing the distribution of scaffold lengths in the *Anas flavirostris* genome assembly.

Assembly pattern:

The genome shows a mixture of very long scaffolds and numerous shorter sequences. The five longest scaffolds range from approximately 75.3 Mb to 201.5 Mb, indicating that a large portion of the assembly is concentrated in a few long sequences, while the high total sequence count of 194,180 shows that many shorter scaffolds are also present.



Part 4 — Perform a Simple Length-Filtering Experiment

Metric	Original genome	After ≥10 kb filter
Total length	1,166,830,609	1,092,679,295
Number of sequences	194,180	385
Maximum length	201,470,136	201,470,136
N50	75,318,589	79,851,999
GC content	41.74	41.31

Answer these questions in 1–2 sentences each:

How many sequences were removed by the 10 kb filter?

- 193,795 sequences were removed by the 10 kb filter, decreasing the number of sequences from 194,180 to 385.

Did the total assembly length decrease greatly or only slightly?

- The total assembly length decreased only slightly, by approximately 74.15 Mb or 6.36% of the original assembly. This indicates that most of the assembled bases were contained in sequences at least 10 kb long.

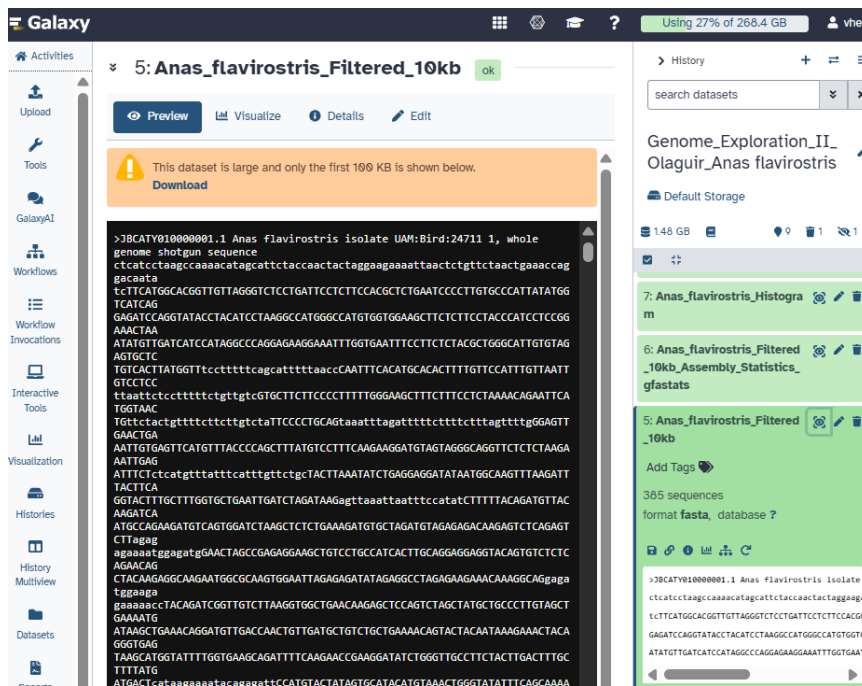


Figure 5. Galaxy output showing the filtered *Anas flavivirus* genome assembly after applying the 10 kb minimum sequence-length filter.

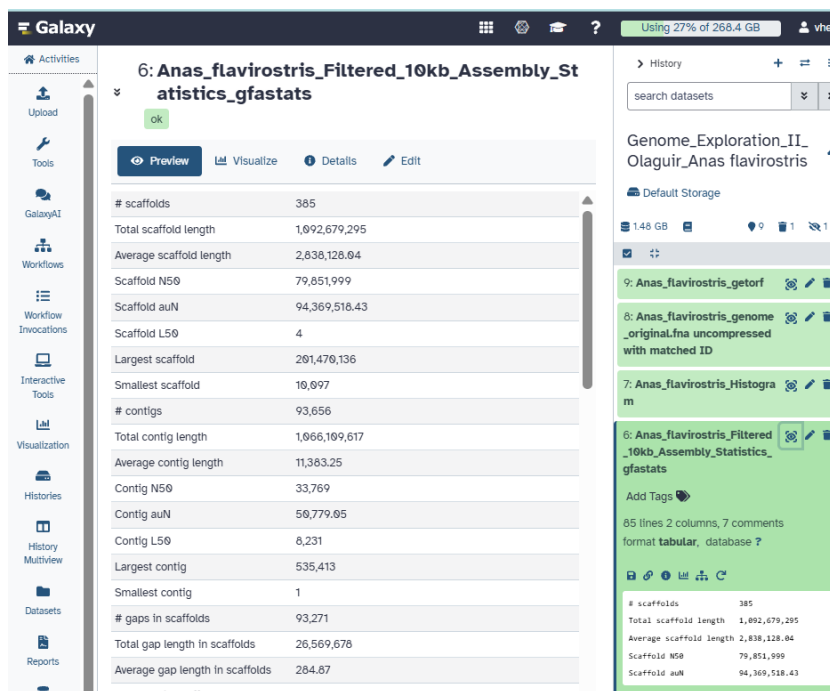


Figure 6. Galaxy gfastats output showing assembly statistics after applying the 10 kb minimum sequence-length filter to the *Anas flavivirus* genome.

Did N50 change? Why do you think it changed or stayed similar?

- The N50 increased from 75,318,589 bp to 79,851,999 bp after filtering. This occurred because many short sequences were removed, leaving a smaller set of longer sequences that contributed more strongly to the 50% assembly-length threshold.

What does this tell you about the contribution of short sequences to your assembly?

- The results show that short sequences were numerous but contributed a relatively small proportion of the total assembly length. Removing them eliminated most of the sequence records while retaining approximately 93.64% of the original assembly length.

Part 5 — Small ORF Exploration

Metric	Value
Selected sequence	JBCATY010000161.1_
Sequence length	100,548 bp
Minimum ORF size used	300 bp
Number of ORFs found	245
Longest ORF	9,069 bp (JBCATY010000161.1_117)

[illegible]

Figure 7. Galaxy output showing the uncompressed *Anas flavirostris* genome sequence used for ORF analysis.

[illegible]

Figure 8. Galaxy getorf output showing the predicted open reading frames (ORFs) from the selected *Anas flavirostris* sequence.

Answer:

How many ORFs were found in the selected sequence/region?

- The uploaded getorf result contains at least 153 ORFs. However, the file is truncated at ORF 153, so the exact total cannot be confirmed from the uploaded result alone.

What was the longest ORF?

- The longest ORF was JBCATY010000161.1_117, spanning 89,449–98,517, with a length of 9,069 bp.

Why should you NOT report every ORF as a real gene?

- An ORF only represents a sequence that can potentially encode a protein; it does not confirm that the sequence is a functional gene. Additional evidence such as genome annotation, transcript data, protein similarity, or gene-prediction analysis is needed to confirm a gene.

Part 6 — Short Biological Interpretation

The genome assembly of *Anas flavirostris* contains a total of 1,166,830,609 bp distributed across 194,180 sequences, with a maximum scaffold length of 201,470,136 bp. The assembly contains several very long scaffolds, as shown by the five longest sequences ranging from approximately 75.3 Mb to 201.5 Mb. The N50 of 75,318,589 bp and L50 of 5 indicate that at least half of the assembled genome is concentrated within only five long scaffolds. However, the large number of sequences also indicates that many shorter scaffolds are present. The 10 kb filtering experiment removed 193,795 sequences, but approximately 93.64% of the original assembly length remained, showing that the short sequences were numerous but contributed a relatively small proportion of the total assembly length. The N50 increased to 79,851,999 bp after filtering because the remaining sequences were predominantly longer. The original assembly has a GC content of 41.74%. The ORF analysis demonstrated that genomic DNA can contain sequence regions with potential protein-coding capacity; however, an ORF alone does not establish that a functional gene is present. Additional annotation, expression, or protein-similarity evidence is needed to confirm a gene.

Part 7 — Galaxy Reproducibility and GitHub Documentation

GALAXY

Student's Name: Olaguir, Junavhel Jane B.

Email: olaguirjunajane@gmail.com

Username: vhel

Link:

<https://usegalaxy.org/u/vhel/h/genome-exploration-ii-olaguir-anas-flavirostris>

GITHUB

Student's Name: Olaguir, Junavhel Jane B.

Email: olaguirjunajane@gmail.com

Username: vhellanine

Link:

https://github.com/vhellanine/OLAGUIR_assignment_02_genome_exploration

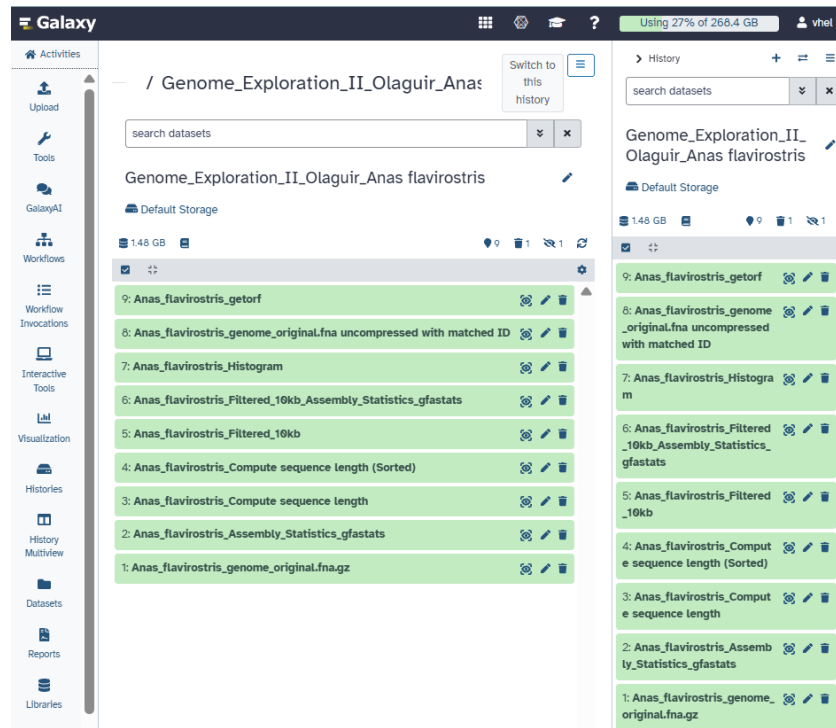


Figure 9. Galaxy history showing the completed workflow and generated datasets for the *Anas flavirostris* genome exploration.

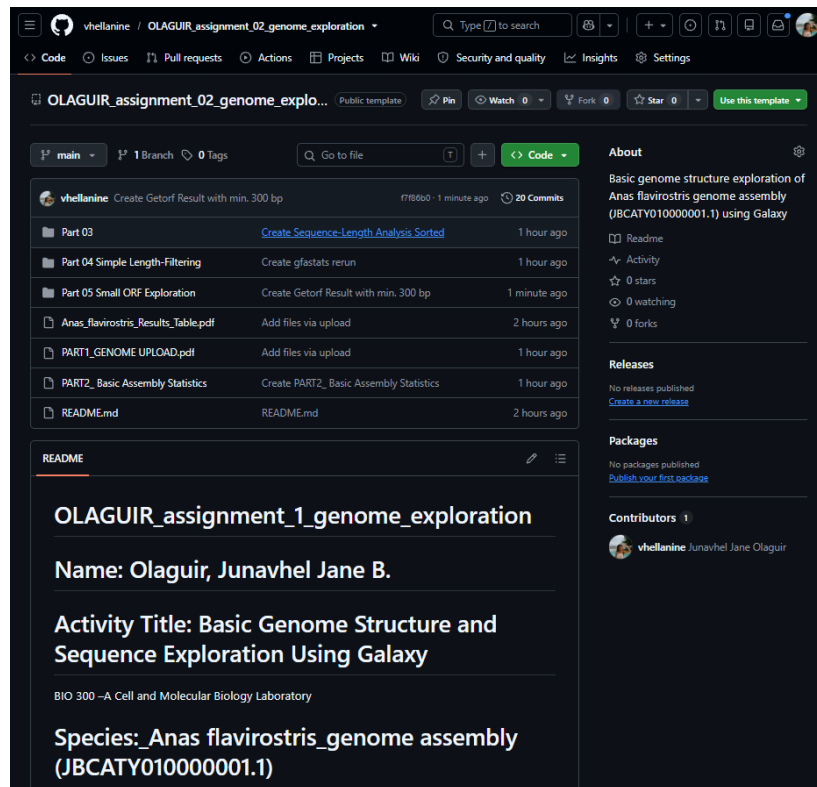


Figure 10. GitHub repository showing the uploaded *Anas flavirostris* genome exploration outputs, including the results table, activity parts, and README documentation.